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deripla

SI2500 / SI3300 / SI5000

Bit	Alarms	01
08	DC1 startup isolation failure	
07	reserved	
06	DC1 operate isolation failure	
05	reserved	
04	reserved	
03	reserved	
02	reserved	
01	reserved	

Bit	Status DC	02
08	Internal DC hardware failure	
07	Internal communication error	
06	Internal bulk failure	
05	Solar power too low	
04	Temperature limited operation	
03	Power limited operation	
02	MPP limited operation	
01	Limits DC	

Bit	Limits DC	03
08	DC1 too low	
07	DC1 critical undervoltage	
06	DC1 undervoltage	
05	DC1 overvoltage	
04	reserved	
03	reserved	
02	reserved	
01	reserved	

Bit	Status AC	04
08	Internal AC hardware failure	
07	Internal ENS hardware failure	
06	Internal AC hardware disturbance	
05	reserved	
04	Normal operation	
03	Sync to AC	
02	reserved	
01	Limits AC	

Bit	Limits AC	05
08	Critical undervoltage	
07	Undervoltage	
06	Overvoltage	
05	Critical Overvoltage	
04	Low frequency	
03	High frequency	
02	Islanding	
01	DC injection current	

Bit	Isolation Warning Status	06
08	reserved	
07	Start-up input isolation Status	
06	PV+ grounded status	
05	Running input isolation Status	
04	PV- grounded status	
03	reserved	
02	Varistor Status	
01	reserved	

Bit	DC Hardware Error	07
08	NTC defect	
07	NTC overtemperature	
06	reserved	
05	reserved	
04	MOV failure	
03	reserved	
02	reserved	
01	ADC error	

Bit	AC Hardware Error	08
08	NTC defect	
07	NTC overtemperature	
06	reserved	
05	Helpingvoltage failure	
04	Overcurrent shutdown	
03	Calibration data corrupted	
02	reserved	
01	Max. AC power shutdown	

alarms				
priority	send alarm	display	Changes	notes
			00	
08	no	yes		
07	no	yes		
06	no	yes		
05	no	yes		
04	no	yes		
03	no	yes		
02	no	yes		
01	no	yes		
			40	
08	no	yes		
07	yes	yes		
06	no	yes		
05	no	yes		
04	no	yes		
03	no	yes		
02	no	yes		
01	no	yes		
			10	
08	no	yes		
07	no	yes		
06	no	yes		
05	yes	yes		
04	no	yes		
03	no	yes		
02	no	yes		
01	no	yes		
			C0	
08	yes	yes		
07	yes	yes		
06	no	yes		
01	no	yes		
05	no	yes		
04	no	yes		
02	no	yes		
03	no	yes		
			FD	
06	yes	yes		
05	yes	yes		
08	yes	yes		
07	yes	yes		
04	yes	yes		
03	yes	yes		
02	no	yes		
01	yes	yes		
			00	nicht in SI2500old und SI3300old
08	no	yes		
07	no	yes		
06	no	yes		
05	no	yes		
04	no	yes		
03	no	yes		
02	no	yes		
01	no	yes		
			C1	
08	yes	yes		
07	yes	yes		
06	no	yes		
05	no	yes		
04	no	yes		
03	no	yes		
02	no	yes		
01	yes	yes		nur SI5000
			CC	
06	yes	yes		
05	yes	yes		
02	no	yes		
03	no	yes		
08	yes	yes		
07	yes	yes		
01	no	yes		
04	no	yes		

Bit	ENS Hardware Error
08	Hardware error
07	reserved
06	reserved
05	reserved
04	reserved
03	reserved
02	reserved
01	reserved

09

Bit	Bulk Error
08	Undervoltage
07	Overvoltage
06	reserved
05	reserved
04	reserved
03	reserved
02	reserved
01	reserved

10

Bit	Communication error
08	AC to DC failure
07	AC to ENS failure
06	AC to Display failure
05	reserved
04	AC to DC error
03	AC to ENS error
02	AC to Display error
01	reserved

11

Bit	AC Hardware Disturbances
08	RTC error
07	AC Relay error
06	reserved
05	reserved
04	reserved
03	reserved
02	reserved
01	reserved

12

			00	
08	no	yes		
07	no	yes		
06	no	yes		
05	no	yes		
04	no	yes		
03	no	yes		
02	no	yes		
01	no	yes		
			00	
08	no	yes		
07	no	yes		
06	no	yes		
05	no	yes		
04	no	yes		
03	no	yes		
02	no	yes		
01	no	yes		
			EE	
08	yes	yes		
07	yes	yes		
06	yes	yes		
05	no	yes		
04	yes	yes		
03	yes	yes		
02	yes	yes		
01	no	yes		
			C0	nicht in SI2500old und SI3300old
08	yes	yes		
07	yes	yes		
06	no	yes		
05	no	yes		
04	no	yes		
03	no	yes		
02	no	yes		
01	no	yes		

send a alarm: if the Error condition is stabil for 10 minutes
send the alarm once per day.

Bit	Status System (1)
08	Isolation error String 2
07	Isolation error String 1
06	reserved
05	Varistor DC damaged
04	Varistor AC damaged
03	Quint3 20A
02	Quint3 3.5A
01	BPC communication error

Bit	Status System (2)
08	reserved
07	reserved
06	GUI OFF
05	Temp. Sensor damaged
04	fan 3 blocked
03	fan 2 blocked
02	fan 1 blocked
01	PowerControl

Bit	Status System (3)
08	reserved
07	reserved
06	reserved
05	reserved
04	reserved
03	reserved
02	date automatically set @start-up
01	BPC not responding @start-up

Bit	AC Hardware Errors
08	NTC defect
07	NTC overtemperature
06	HW revision mismatch
05	Helpingvoltage
04	Overcurrent shutdown
03	Calibration data corrupted
02	Bulk under voltage
01	Bulk over voltage

Bit	AC Comm. Errors
08	Ext. Comm. timeout
07	IO fail
06	DC fail
05	ENS fail
04	IO broken
03	DC broken
02	ENS broken
01	reserved

Bit	AC GRID ERRORS
08	Critical undervoltage
07	Undervoltage
06	Overvoltage
05	Critical overvoltage
04	Low frequency
03	High frequency
02	reserved
01	DC injection current

Bit	Calibration errors
08	reserved
07	reserved
06	DC HW revision error
05	AC HW revision error
04	Relay error
03	ENS compensation error
02	AC compensation error
01	LEM calibration error

Bit	DC Hardware Error
08	NTC defect
07	NTC overtemperature
06	ENS_OK error
05	AC_OK error
04	reserved
03	Hardware error
02	Buck error
01	Bridge error

alarms					
priority	send alarm	display			
08	no	yes			
07	no	yes			
04	no	yes			
03	yes	yes			
02	yes	yes			
01	yes	yes			
06	yes	yes			
05	no	yes			
08	no	yes			
07	no	yes			
06	yes	yes			
05	yes	yes			
04	yes	yes			
03	yes	yes			
02	yes	yes			
01	no	yes			
08	no	yes			
07	no	yes			
06	no	yes			
05	no	yes			
04	no	yes			
03	no	yes			
02	yes	yes		new	
01	yes	yes		new	
04/12/20/28/36/44/52/60/68					
08	yes	yes			
07	yes	yes			
06	no	yes			
05	no	yes			
04	yes	yes			
03	yes	yes			
02	no	yes			
01	no	yes			
05/13/21/29/37/45/53/61/69					
07	no	yes			
06	yes	yes			
05	yes	yes			
04	yes	yes			
03	yes	yes			
02	yes	yes			
01	yes	yes			
08	no	yes			
06/14/22/30/38/46/54/62/70					
08	yes	yes			
07	yes	yes			
06	yes	yes			
05	yes	yes			
04	yes	yes			
03	yes	yes			
02	no	yes			
01	yes	yes			
07/15/23/31/39/47/55/63/71					
08	no	yes			
07	no	yes			
06	no	yes			
05	yes	yes			
04	yes	yes			
03	yes	yes			
02	yes	yes			
01	yes	yes			
08/16/24/32/40/48/56/64/72					
08	yes	yes			
07	yes	yes			
06	no	yes			
05	no	yes			
04	no	yes			
03	yes	yes			
02	yes	yes			
01	yes	yes			

Bit	DC Voltage error	09/17/25/33/41/49/57/65/73				
08	reserved	08	no	yes		
07	reserved	07	no	yes		
06	reserved	06	no	yes		
05	reserved	05	no	yes		
04	reserved	04	no	yes		
03	Input Over voltage	03	yes	yes		
02	input voltage start up	02	no	yes		
01	Input Under voltage	01	no	yes		
Bit	IO TWI comm status	10/18/26/34/42/50/58/66/74				
08	reserved	08	no	yes		
07	reserved	07	no	yes		
06	reserved	06	no	yes		
05	reserved	05	no	yes		
04	reserved	04	no	yes		
03	reserved	03	no	yes		
02	reserved	02	no	yes		
01	error	01	yes	yes		
Bit	External Error	11/19/27/35/43/51/59/67/75				
08	reserved	08	no	yes		
07	reserved	07	no	yes		
06	reserved	06	no	yes		
05	reserved	05	no	yes		
04	Power unsymmetric	04	yes	yes	new	
03	I/O discrepancy	03	yes	yes	new	
02	Phase not responding	02	no	yes		
01	Modul not responding	01	no	yes		
Bit	Status byte	76/77/78/79/80/81/82/83/84				
08	Master of BUS	08	no	no		
07	Master of MPPT1	07	no	no		
06	Master of MPPT2	06	no	no		
05	Off	05	no	no		
04	Error	04	no	no		
03	Working	03	no	no		
02	Ready	02	no	no		
01	Present	01	no	no		

send a alarm: if the Error condition is stabil for 10 minutes
send the alarm once per day.

Solivia G3

Bit	Alarms
08	Startup isolation failure
07	reserved
06	Running isolation failure
05	reserved
04	GFDI Hardware error
03	Fuse plus failure
02	Fuse minus failure
01	GFDI test failure

01

Bit	Status DC
08	Internal DC hardware failure
07	Internal communication error
06	Internal bulk failure
05	Input power too low
04	Temperature limited operation
03	Power limited operation
02	MPP limited operation
01	Limits DC

02

Bit	Limits DC
08	Input voltage too low
07	Input critical undervoltage
06	reserved
05	reserved
04	reserved
03	reserved
02	reserved
01	reserved

03

Bit	Status System Ctrl (SC)
08	Internal AC hardware failure
07	Internal SC hardware failure
06	Internal AC hardware disturbance
05	reserved
04	Normal operation
03	Sync to Grid on going
02	Calibration on going
01	Grid errors

04

Bit	Grid errors
08	Critical undervoltage
07	Undervoltage
06	Overvoltage
05	Critical Overvoltage
04	Low frequency
03	High frequency
02	Islanding
01	DC injection current

05

Bit	Isolation Warning Status
08	reserved
07	Start up isolation Status
06	PV+ grounded status
05	Running isolation Status
04	PV- grounded status
03	reserved
02	Varistor Status
01	reserved

06

Bit	DC Hardware Error
08	NTC defect
07	NTC overtemperature
06	SC ok error
05	reserved
04	reserved
03	reserved
02	Boost error
01	Bridge error

07

Bit	AC Hardware Error
08	NTC defect
07	NTC overtemperature
06	HW bulk undervoltage
05	Aux voltage
04	Overcurrent shutdown
03	Eeprom corrupted
02	Bulk overvoltage
01	Bulk undervoltage

08

alarms				
priority	send alarm	display	hex	notes
			00	
08	no	yes		nur EU Geräte
07	no	yes		
06	no	yes		nur EU Geräte
05	no	yes		
04	yes	yes		nur US Geräte
03	yes	yes		nur US Geräte
02	yes	yes		nur US Geräte
01	no	yes		nur US Geräte
			40	
08	no	yes		
07	yes	yes		
06	no	yes		
05	no	yes		
04	no	yes		
03	no	yes		
02	no	yes		
01	no	yes		
			00	
08	no	yes		
07	no	yes		
06	no	yes		
05	no	yes		
04	no	yes		
03	no	yes		
02	no	yes		
01	no	yes		
			C0	
08	yes	yes		
07	yes	yes		
06	no	yes		
01	no	yes		
05	no	yes		
04	no	yes		
02	no	yes		
03	no	yes		
			FD	
06	yes	yes		
05	yes	yes		
08	yes	yes		
07	yes	yes		
04	yes	yes		
03	yes	yes		
02	no	yes		
01	yes	yes		
			00	
08	no	yes		
07	no	yes		nur EU Geräte
06	no	yes		nur EU Geräte
05	no	yes		nur EU Geräte
04	no	yes		nur EU Geräte
03	no	yes		
02	no	yes		
01	no	yes		
			C0	
08	yes	yes		
07	yes	yes		
06	no	yes		
05	no	yes		
04	no	yes		
03	no	yes		
02	no	yes		
01	no	yes		
			CC	
06	yes	yes		
05	yes	yes		
02	no	yes		
03	no	yes		
08	yes	yes		
07	yes	yes		
01	no	yes		
04	no	yes		

Bit	SC Hardware Error	09
08	AC OK ERROR	
07	reserved	
06	reserved	
05	reserved	
04	reserved	
03	reserved	
02	reserved	
01	reserved	

Bit	Bulk Error	10
08	Undervoltage	
07	Overvoltage	
06	reserved	
05	reserved	
04	reserved	
03	reserved	
02	reserved	
01	reserved	

Bit	Communication error	11
08	DC failure	
07	AC failure	
06	reserved	
05	LCD failure	
04	DC error	
03	AC error	
02	LCD error	
01	reserved	

Bit	SC Hardware Disturbances	12
08	RTC error	
07	AC Relay error	
06	reserved	
05	reserved	
04	reserved	
03	reserved	
02	IT autotest fail	
01	Ext. Power Control active	

Bit	DC HW stage error	13
08	reserved	
07	reserved	
06	reserved	
05	reserved	
04	Overvoltage 2	
03	Current protection	
02	Aux voltage error	
01	Overvoltage	

Bit	Calibration status	14
08	reserved	
07	reserved	
06	reserved	
05	reserved	
04	reserved	
03	SC calibration error	
02	AC calibration error	
01	LEM calibration error	

Bit	Neutral errors	15
08	Critical undervoltage N1	
07	Undervoltage N1	
06	Overvoltage N1	
05	Critical Overvoltage N1	
04	Critical undervoltage N2	
03	Undervoltage N2	
02	Overvoltage N2	
01	Critical Overvoltage N2	

			00	
08	no	yes		
07	no	yes		
06	no	yes		
05	no	yes		
04	no	yes		
03	no	yes		
02	no	yes		
01	no	yes		
			00	
08	no	yes		
07	no	yes		
06	no	yes		
05	no	yes		
04	no	yes		
03	no	yes		
02	no	yes		
01	no	yes		
			DE	
08	yes	yes		
07	yes	yes		
06	no	yes		
05	yes	yes		
04	yes	yes		
03	yes	yes		
02	yes	yes		
01	no	yes		
			C1	
08	yes	yes		
07	yes	yes		
06	no	yes		
05	no	yes		
04	no	yes		
03	no	yes		
02	no	yes		
01	yes	yes		
			00	
08	no	yes		
07	no	yes		
06	no	yes		
05	no	yes		
04	no	yes		
03	no	yes		
02	no	yes		
01	no	yes		
			00	
08	no	yes		
07	no	yes		
06	no	yes		
05	no	yes		
04	no	yes		
03	no	yes		
02	no	yes		
01	no	yes		
			99	nur US Geräte
04	yes	yes		
03	no	yes		
08	no	yes		
07	yes	yes		
02	yes	yes		
01	no	yes		
06	no	yes		
05	yes	yes		

send a alarm: if the Error condition is stabil for 10 minutes
send the alarm once per day.

Delta Solivia 15/20/30 TL

		Bit	Alarms	Error Code
Status AC Output 1		0	reserved	
64	1	1	Over Frequency Range	E01
		2	Under Frequency Range	E02
		3	reserved	E03
		4	reserved	E04
		5	reserved	E05
		6	Phase Jump	E06
		7	Grid Quality	E07
Status AC Output 2		0	Ac connected Fail	E08
64	2	1	No Grid	E09
		2	Under Voltage Range_R	E10
		3	Over Voltage Range_R	E11
		4	Over Transient Voltage Range_R	E12
		5	Slow Over Voltage Range_R	E13
		6	reserved	E14
		7	Under Voltage Range_S	E15
Status AC Output 3		0	Over Voltage Range_S	E16
64	3	1	Over Transient Voltage Range_S	E17
		2	Slow Over Voltage Range_S	E18
		3	reserved	E19
		4	Under Voltage Range_T	E20
		5	Over Voltage Range_T	E21
		6	Over Transient Voltage Range_T	E22
		7	Slow Over Voltage Range_T	E23
Status AC Output 4		0	reserved	E24
64	4	1	3 Phases Unbalance	E25
		2	reserved	E26
		3	reserved	E27
		4	reserved	E28
		5	reserved	
		6	reserved	
		7	reserved	
Status DC Input 1		0	reserved	
64	5	1	reserved	
		2	reserved	
		3	reserved	
		4	reserved	
		5	reserved	E29
		6	String1 PV input voltage too high	E30
		7	String2 PV input voltage too high	E31
Status DC Input 2		0	reserved	
64	6	1	SOLAR_1_UVR	W01
		2	SOLAR_2_UVR	W02
		3	reserved	W03
		4	reserved	W04
		5	reserved	W05
		6	reserved	W06
		7	reserved	W07
Error Status		0	reserved	W08
65	1	1	reserved	W09
		2	reserved	W10
		3	FAN_FAIL	W11
		4	reserved	W12
		5	reserved	W13
		6	reserved	W14
		7	reserved	W15
Error Status AC 1		0	reserved	
65	2	1	DC Injection_R	F01
		2	DC Injection_S	F02
		3	DC Injection_T	F03
		4	DC Injection_Total	F04
		5	Inside OTP	F05
		6	Inside NTC Circuit Fail	F06
		7	Heat Sink over temperature	F07

alarms		
priority	send alarm	display
01	no	yes
02	yes	yes
03	yes	yes
04	no	yes
05	no	yes
06	no	yes
07	no	yes
08	yes	yes
01	yes	yes
02	yes	yes
03	yes	yes
04	yes	yes
05	yes	yes
06	yes	yes
07	no	yes
08	yes	yes
01	yes	yes
02	yes	yes
03	yes	yes
04	no	yes
05	yes	yes
06	yes	yes
07	yes	yes
08	yes	yes
01	no	yes
02	no	yes
03	no	yes
04	no	yes
05	no	yes
06	no	yes
07	no	yes
08	no	yes
01	no	yes
02	no	yes
03	no	yes
04	no	yes
05	no	yes
06	no	yes
07	yes	yes
08	yes	yes
01		
02	yes	yes
03	yes	yes
04	no	yes
05	no	yes
06	no	yes
07	no	yes
08	no	yes
01	no	yes
02	no	yes
03	no	yes
04	yes	yes
05	no	yes
06	no	yes
07	no	yes
08	no	yes
01	no	yes
02	no	yes
03	no	yes
04	no	yes
05	yes	yes
06	no	yes
07	yes	yes
08	yes	yes

Global Error 1		0	Heat Sink Ntc1 Circuit Fail	F08
65	3	1	Heat Sink Ntc2 Circuit Fail	F09
		2	Heat Sink Ntc3 Circuit Fail	F10
		3	Inverter Choke over temperature	F11
		4	reserved	F12
		5	Relay Open	F13
		6	Firmware Incompatibility	F14
		7	DSP ADC Vgrid/Iout Bias Fail	F15

CPU Error		0	DSP ADC Vin/Vbus Bias Fail	F16
65	4	1	DSP ADC Iin/Iboost Bias Fail	F17
		2	Red. ADC Vgrid/Vinv Bias Fail	F18
		3	Red. ADC Iout_dc Bias Fail	F19
		4	Efficiency Abnormal	F20
		5	Fan Fail	F21
		6	Internal Communication Fault (between Redundant)	F22
		7	Internal Communication Fault (between Display)	F23

Global Error 2		0	Residual Current Over Rating	F24
65	5	1	Insulation	F25
		2	3P Current Unbalance	F26
		3	RCMU Circuit Fail	F27
		4	Relay Test Short	F28
		5	Relay Test Open	F29
		6	Bus unbalance	F30
		7	Bus_P over voltage rating	F31

Limits AC output 1		0	Bus_P under voltage rating	F32
66	1	1	Bus_N over voltage rating	F33
		2	Bus_N under voltage rating	F34
		3	Bus voltage over rating	F35
		4	Output Current Transient Over Rating_R	F36
		5	Output Current Over Rating_R	F37
		6	Output Current Transient Over Rating_S	F38
		7	Output Current Over Rating_S	F39

Limits AC output 2		0	Output Current Transient Over Rating_T	F40
66	2	1	Output Current Over Rating_T	F41
		2	CT current sensor Fail_A	F42
		3	CT current sensor Fail_B	F43
		4	CT current sensor Fail_C	F44
		5	HW OSCP circuit	F45
		6	Inverter Failure	F46
		7	reserved	F47

Global Error 3		0	reserved	F48
66	3	1	reserved	F49
		2	Zero Cross Circuit Fail	F50
		3	reserved	F51
		4	reserved	F52
		5	reserved	F53
		6	reserved	F54
		7	reserved	F55

Limits DC 1		0	reserved	F56
66	4	1	reserved	F57
		2	reserved	F58
		3	reserved	F59
		4	PV1 Current Over Rating	F60
		5	PV2 Current Over Rating	F61
		6	reserved	F62
		7	reserved	F63

Limits DC 2		0	power control active	F64
66	5	1	reserved	F65
		2	reserved	F66
		3	reserved	F67
		4	reserved	F68
		5	reserved	F69
		6	PV1 Current Transient Over Rating	F70
		7	PV2 Current Transient Over Rating	F71

01	yes	yes
02	yes	yes
03	yes	yes
04	no	yes
05	no	yes
06	no	yes
07	yes	yes
08	no	yes

01	no	yes
02	no	yes
03	no	yes
04	no	yes
05	no	yes
06	no	yes
07	no	yes
08	no	yes

01	no	yes
02	yes	yes
03	no	yes
04	no	yes
05	no	yes
06	no	yes
07	no	yes
08	no	yes

01	no	yes
02	no	yes
03	no	yes
04	no	yes
05	no	yes
06	no	yes
07	no	yes
08	no	yes

01	no	yes
02	no	yes
03	no	yes
04	no	yes
05	no	yes
06	no	yes
07	no	yes
08	no	yes

01	no	yes
02	no	yes
03	no	yes
04	no	yes
05	no	yes
06	no	yes
07	no	yes
08	no	yes

01	no	yes
02	no	yes
03	no	yes
04	no	yes
05	yes	yes
06	yes	yes
07	no	yes
08	no	yes

01	no	yes
02	no	yes
03	no	yes
04	no	yes
05	no	yes
06	no	yes
07	yes	yes
08	yes	yes

send a alarm: if the Error condition is stabil for 10 minutes
send the alarm once per day.

Delta Solivia TR G4

13.01.2012

Status 1	alarms		
	priority	send alarm	display
bit 0 = 1 -> Self test ongoing	01	no	yes
bit 1 = 1 -> Firmware update	02	no	yes
bit 2 = 1 -> Night mode	03	no	yes
bit 3 = 1 -> L1 Voltage failure	04	no	yes
bit 4 = 1 -> L2 Voltage failure	05	no	yes
bit 5 = 1 -> L3 Voltage failure	06	no	yes
bit 6 = 1 -> L1 Frequency failure	07	no	yes
bit 7 = 1 -> L2 Frequency failure	08	no	yes
bit 8 = 1 -> L3 Frequency failure	09	no	yes
bit 9 = 1 -> L1 DC Inj. failure	10	no	yes
bit 10 = 1 -> L2 DC Inj. Failure	11	no	yes
bit 11 = 1 -> L3 DC Inj. failure	12	no	yes
bit 12 = 1 -> L1 islanding	13	no	yes
bit 13 = 1 -> L2 islanding	14	no	yes
bit 14 = 1 -> L3 islanding	15	no	yes
bit 15 = 1 -> L1 Grid error	16	no	yes
bit 16 = 1 -> L2 Grid error	17	no	yes
bit 17 = 1 -> L3 Grid error	18	no	yes
bit 18 = 1 -> L1 Long grid out	19	no	yes
bit 19 = 1 -> L2 Long grid out	20	no	yes
bit 20 = 1 -> L3 Long grid out	21	no	yes
bit 21 = 1 -> L1 Grid synchronization error	22	no	yes
bit 22 = 1 -> L2 Grid synchronization error	23	no	yes
bit 23 = 1 -> L3 Grid synchronization error	24	no	yes

bit 24..31 RESERVED

Status 3	alarms		
	priority	send alarm	display
bit 0 = 1 -> PV1 Iso startup warning	01	no	yes
bit 1 = 1 -> PV1 Iso running warning	02	no	yes
bit 2 = 1 -> PV1+ grounding warning	03	no	yes
bit 3 = 1 -> PV1- grounding warning	04	no	yes
bit 4 = 1 -> PV2 Iso startup warning	05	no	yes
bit 5 = 1 -> PV2 Iso running warning	06	no	yes
bit 6 = 1 -> PV2+ grounding warning	07	no	yes
bit 7 = 1 -> PV2- grounding warning	08	no	yes
bit 8 = 1 -> PV3 Iso startup warning	09	no	yes
bit 9 = 1 -> PV3 Iso running warning	10	no	yes
bit 10 = 1 -> PV3 + grounding warning	11	no	yes
bit 11 = 1 -> PV3- grounding warning	12	no	yes
bit 12 = 1 -> PV1 voltage too low warning	13	no	yes
bit 13 = 1 -> PV2 voltage too low warning	14	no	yes
bit 14 = 1 -> PV3 voltage too low warning	15	no	yes

bit 15..31 RESERVED

Status 2	alarms		
	priority	send alarm	display
bit 0 = 1 -> PV1 Iso startup failure	01	no	yes
bit 1 = 1 -> PV1 Iso running failure	02	no	yes
bit 2 = 1 -> PV1+ grounding failure	03	no	yes
bit 3 = 1 -> PV1- grounding failure	04	no	yes
bit 4 = 1 -> PV2 Iso startup failure	05	no	yes
bit 5 = 1 -> PV2 Iso running failure	06	no	yes
bit 6 = 1 -> PV2+ grounding failure	07	no	yes
bit 7 = 1 -> PV2- grounding failure	08	no	yes
bit 8 = 1 -> PV3 Iso startup failure	09	no	yes
bit 9 = 1 -> PV3 Iso running failure	10	no	yes
bit 10 = 1 -> PV3+ grounding failure	11	no	yes
bit 11 = 1 -> PV3- grounding failure	12	no	yes
bit 12 = 1 -> PV1 voltage too low failure	13	no	yes
bit 13 = 1 -> PV2 voltage too low failure	14	no	yes
bit 14 = 1 -> PV3 voltage too low failure	15	no	yes
bit 15 = 1 -> Internal failure	16	yes	yes
bit 16 = 1 -> Auto test failure	17	no	yes
bit 17 = 1 -> PV power too low failure	18	no	yes

bit 18..31 RESERVED

Status 4	alarms		
	priority	send alarm	display
bit 0 = 1 -> Internal fan warning	01	no	yes
bit 1 = 1 -> External fan warning	02	no	yes
bit 2 = 1 -> Synchronization ongoing	03	no	yes

bit 3..31 RESERVED